

RETHINKING SLEEP STAGING: Unsupervised cross-laboratory mouse sleep staging with temporal variational autoencoders and substage discovery

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ABSTRACT

Cross-laboratory manual scoring of mouse sleep from electroencephalography (EEG) and electromyography (EMG) lacks consensus, and supervised models trained at one site often fail elsewhere without retraining or per-animal calibration. We study twenty quality-screened wild-type mice from three European Mouse Sleep Staging Validation (MSSV) laboratories under leave-mice-out cross-validation with four folds. Decoder-only conditional variational autoencoders (cVAE) assign states to held-out animals from spectral EEG and EMG alone, using only the encoder and a learned mixture prior at inference. A hidden Markov model–Gaussian mixture model (HMM–GMM) prior that models bout structure reached mean prior normalized mutual information (NMI) of 0.575 on joint cross-laboratory holdout, versus 0.282 for a matched static mixture, a gain of 0.29 NMI averaged over folds. Beyond three macro states, holdout macro NMI falls as mixture count K increases while prior log-likelihood improves. We therefore select K for biological interpretability near six to nine substates rather than from macro NMI alone. A population sweep on all twenty mice supports a $K=6$ map with six active substates and distinct Wake and non-rapid eye movement (NREM) components. The method enables comparison of sleep substage structure across acquisition sites before expert labels are harmonised.

AUTHOR SUMMARY

Sleep biologists score mouse EEG and EMG into Wake, NREM, and rapid eye movement (REM) sleep, but raters at different laboratories often disagree. We trained unsupervised models on mice from Bern, Copenhagen, and Lyon, then tested them on held-out animals the models had never seen, without calibrating each new recording by hand. Our best model reads spectral brain and muscle signals through a neural network and enforces realistic sleep bout structure with a temporal prior. It outperformed simpler baselines on cross-laboratory holdout and recovered finer substates whose spectra and bout lengths matched patterns we discussed with sleep neuroscientists at the Kornum laboratory. The approach is not yet accurate enough to replace supervised scoring, especially for REM, but it shows that substage structure can be studied across sites before expert labels are aligned.

1 INTRODUCTION

Manual mouse sleep scores disagree across MSSV laboratories, and supervised models trained at one site fail on held-out animals elsewhere unless retrained or calibrated per mouse [1–5]. Expert staging nonetheless maps continuous physiology into Wake, NREM, and REM, the standard three-class benchmark. Unsupervised work in mice has repeatedly resolved substages with distinct spectra, bout lengths, and transition roles that Wake/NREM/REM labels alone do not name [6–8]. Montage and laboratory protocols differ across studies [9]. Claims that sleep structure extends beyond macro scoring must therefore be tested on new animals and new sites under population holdout without per-animal calibration at inference.

Unsupervised sequence models infer latent structure directly from EEG and EMG dynamics and align naturally with bout persistence and transitions [7]. Static clusterers and shallow hidden Markov models

(HMMs) discover states from raw or hand-crafted features but rarely evaluate zero-shot transfer to held-out mice from other laboratories under a shared population model [10–12]. Katsageorgiou et al. resolved rodent substages beyond the macro taxonomy with distinct physiology and bout structure in mcRBM [6], yet without cross-laboratory leave-mice-out holdout. García Ciudad et al. showed that longer temporal context improves supervised rodent staging, especially for REM [13]. That setting remains fully supervised and does not test unsupervised cross-site transfer. In a single-laboratory cohort with encoder+decoder conditioning and expert features, mixture-prior conditional Gaussian mixture variational autoencoder (cGMVAE) reached population prior NMI of about 0.70 under leave-one-subject-out prediction [8]. Cross-laboratory holdout on spectral convolutional neural network (CNN) inputs is substantially harder and is the focus here.

We extend decoder-only cVAE with HMM–GMM mixture priors [14–17]. Subject embeddings condition reconstruction during training only. Held-out mice are staged from encoder means and the learned prior without subject identifiers. We validate zero-shot cross-laboratory holdout at $K=3$, then test whether latents support finer structure than Wake, NREM, and REM through mixture-resolution sweeps and physiology-first interpretation [6, 8]. We compare a static Gaussian mixture model (GMM) prior with a warm conditional HMM–GMM variational autoencoder (cHMM–GMVAE) prior on locked spectral inputs across laboratories 2, 3, and 5 at Bern, Copenhagen, and Lyon. Holdout fold 4 is the generalisation benchmark for macro metrics. The full cohort supports substage physiology panels in the Appendix. Dataset and protocol details are in Section 2 and Appendix A.

2 DATA

Dataset and cohort. MSSV on OpenNeuro ds006366 [9] comprises invasive EEG and neck EMG from 92 mice across five European laboratories. Expert Wake, NREM, REM, and Artifact labels are provided on a 4s grid for evaluation only. We analyse laboratories 2 at Bern, 3 at Copenhagen, and 5 at Lyon. These are healthy wild-type sites with different electrode layouts. We exclude laboratories 1 and 4 because they are pathology-specific or single-electrode cohorts. Quality screening retains mice with low EEG drift, defined as relative 0.5–2 Hz power ≤ 0.2 of 0.5–30 Hz on each retained electrode. EMG must separate Wake from NREM with a Wake/NREM root-mean-square ratio ≥ 2 . The screened cohort comprises 20 mice with 6, 10, and 4 per site and ~ 453 h of recording under four-fold leave-mice-out cross-validation. The subset balances cross-laboratory coverage with stable spectral inputs. Training mice never appear in the validation fold for a given split. No new animal procedures were performed.

Preprocessing and inputs. Montages are harmonised to a parietal–frontal EEG differential plus neck EMG. Laboratory 2 uses EEG1 and EEG4. Laboratories 3 and 5 use EEG1 and EEG2. Per-site spectral filters and band limits were fixed from incohort ablations before holdout. Recordings are segmented into 4s epochs at 128 Hz. Artifact epochs are removed. Each epoch is a log-power spectral map $\mathbf{x}_t \in \mathbb{R}^{C \times F}$, batched into contiguous sequences for temporal models without hand-crafted expert features.

3 METHODS

3.1 MODEL AND TRAINING

Spectral EEG/EMG epochs are mapped to a low-dimensional latent space by a cVAE. A subject embedding conditions reconstruction during training only. At holdout, staging uses encoder means and a learned mixture prior without subject identifiers. We compare two prior structures on matched inputs and cross-validation folds. cGMVAE uses a static GMM with $T=1$. cHMM–GMVAE uses a warm HMM–GMM sequence prior with $T=64$ epochs, about 4.3 min, and $K=3$ for macro staging. HMM–GMM variational autoencoder without subject conditioning (HMMGMVAE) provides a sequence-prior baseline in Table 1 [17].

Encoder-decoder. Training batches are contiguous windows $\mathbf{x}_{1:T} \in \mathbb{R}^{T \times C \times F}$ with $T = 64$. A convolutional encoder maps each epoch to a diagonal Gaussian posterior,

$$q_{\phi}(\mathbf{z}_t | \mathbf{x}_t) = \mathcal{N}(\mathbf{z}_t | \boldsymbol{\mu}_t, \text{diag}(\boldsymbol{\sigma}_t^2)), \quad (\boldsymbol{\mu}_t, \log \boldsymbol{\sigma}_t^2) = f_{\phi}(\mathbf{x}_t), \quad (1)$$

with f_{ϕ} a CNN–multilayer perceptron (MLP) encoder and $\mathbf{z}_t \in \mathbb{R}^d$, $d=6$. Latent samples use the reparameterisation $\mathbf{z}_t = \boldsymbol{\mu}_t + \boldsymbol{\sigma}_t \odot \boldsymbol{\epsilon}_t$ with $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The decoder reconstructs each epoch from $\mathbf{z}_t$ and, during

training only, a subject embedding $\mathbf{e}_i \in \mathbb{R}^8$:

$$\hat{\mathbf{x}}_t = g_{\theta}(\mathbf{z}_t, \mathbf{e}_i). \quad (2)$$

Under decoder-only conditioning, f_{ϕ} does not receive $\mathbf{e}_i$. CNN and MLP widths and the full architecture diagram are in Appendix B.1 and Model architecture.

Training objective. For each window $\mathbf{x}_{1:T}$, training minimises

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta \mathcal{L}_{\text{KL}}, \quad (3)$$

where

$$\mathcal{L}_{\text{recon}} = \frac{1}{T} \sum_{t=1}^T \|\mathbf{x}_t - \hat{\mathbf{x}}_t\|_2^2 \quad (4)$$

is mean squared reconstruction error over T contiguous 4s epochs. The Kullback–Leibler (KL) term is the sequence-averaged log-density ratio between the encoder posterior and the mixture prior,

$$\mathcal{L}_{\text{KL}} = \frac{1}{T} \sum_{t=1}^T [\log q_{\phi}(\mathbf{z}_t | \mathbf{x}_t) - \log p(\mathbf{z}_{1:T})]. \quad (5)$$

For cHMM–GMVAE, $\log p(\mathbf{z}_{1:T})$ is the HMM–GMM forward marginal with emission densities $b_k(\mathbf{z}_t)$ and transition matrix A . For cGMVAE, each epoch is scored independently with a static GMM marginal at $T=1$. The weight β is zero for the first ten epochs, then ramps linearly from 0.01 to 1.0. cHMM–GMVAE warms the mixture prior from encoder means [16, 17]. Phase timings and prior equations are in Appendix C.1 and Appendix B.4.

Prior-based staging at holdout. Held-out mice are staged from encoder means $\boldsymbol{\mu}_t$ and the trained prior without subject embeddings or decoder reconstruction. Static cGMVAE assigns $\hat{s}_t = \arg \max_k [\log \pi_k + \log b_k(\boldsymbol{\mu}_t)]$. cHMM–GMVAE applies Viterbi decoding,

$$\hat{\mathbf{s}}_{1:T} = \arg \max_{\mathbf{s}_{1:T}} \left[\log \pi_{s_1} + \sum_{t=2}^T \log A_{s_{t-1}, s_t} + \sum_{t=1}^T \log b_{s_t}(\boldsymbol{\mu}_t) \right], \quad (6)$$

with $B_{tk} = \log b_k(\boldsymbol{\mu}_t)$.

3.2 EVALUATION PROTOCOL

The twenty quality-screened mice were partitioned into four leave-mice-out folds (Appendix A.4). Joint scope pools all non-held-out mice from laboratories 2, 3, and 5 into one population model and tests on the held-out fold across sites. Within-lab scope restricts both training and validation to a single laboratory with the same fold assignments. Held-out mice are staged from encoder means and the trained mixture prior only, as in Eqs. (6) and the holdout paragraph above. Per-laboratory spectral locks remain fixed. Agreement with expert Wake, NREM, and REM is prior NMI,

$$\text{NMI}(Y, \hat{Y}) = \frac{2I(Y; \hat{Y})}{H(Y) + H(\hat{Y})}, \quad (7)$$

where I is mutual information and H Shannon entropy. Each fold trains three seeds; holdout prior NMI uses the best unsupervised checkpoint per seed and reports the maximum across seeds (Appendix B). Macro-aligned accuracy, Cohen’s κ , decoder-only ablation, and substage sweep setup are in Appendix B and C.

Data and code availability. MSSV is available on OpenNeuro as ds006366. Processed features, analysis code, and full experimental settings are provided as supplementary material in Supplementary experimental settings.

4 RESULTS AND DISCUSSION

4.1 CROSS-LABORATORY HOLDOUT AT MACRO RESOLUTION

We evaluated decoder-only cGMVAE and cHMM-GMVAE under joint leave-mice-out training on twenty quality-screened mice from Bern, Copenhagen, and Lyon (Table 1). Supervised pipelines on MSSV reach much higher accuracy when retrained per site or given long temporal context [2, 3, 13]. Under the same unsupervised constraint, cHMM-GMVAE reached mean prior NMI 0.575 versus 0.282 for matched cGMVAE, a gain of 0.29 averaged over four folds.

Table 1. Joint holdout prior NMI by fold (best-of-three seeds, locked recipes, $K=3$). The raw time-domain hidden Markov model (HMM) baseline folds 1–3 report best logged validation prior NMI from incomplete joint runs with ≤ 0.003 . Fold 4 uses the full holdout run.

Fold	HMM (raw)	cGMVAE	HMMGMVAE	cHMM-GMVAE
1	0.003	0.267	0.409	0.579
2	0.001	0.256	0.236	0.396
3	0.003	0.307	0.497	0.675
4	0.208	0.299	0.404	0.649
Mean	0.054	0.282	0.387	0.575

In Table 1, the temporal prior improves every fold. The largest margins appear on folds 3 and 4 at +0.36 and +0.41 NMI. The smallest gain is on fold 2 at +0.10, where Bern montages mix with held-out mice from all sites. Fold 4 confusion (S3 Fig) shows strong Wake and NREM alignment but REM recall near 3%. Thus unsupervised models recover coarse Wake/NREM/REM structure on held-out mice across laboratories, well above chance but below supervised ceilings. Three macro classes remain an incomplete description of the signal.

4.2 LABORATORY-SPECIFIC TRANSFER

Table 2 compares joint population training with within-laboratory training on the *same* held-out mice from each site. For joint training, NMI and macro-aligned accuracy are computed on laboratory-specific holdout subsets only, not pooled across sites.

Table 2. Fair per-laboratory holdout comparison (mean over four folds, $K=3$). Joint training uses all laboratories but reports metrics on held-out mice from the listed site only. Within training trains and evaluates inside the same site. All rows use locked preprocessing per laboratory and matched CNN-variational autoencoder (VAE) capacity with $d=6$, MLP widths 512–256–128, 200 epochs, batch 128, and three seeds. cGMVAE, HMMGMVAE, and cHMM-GMVAE share learning rate 1.3×10^{-3} and differ in embedding use, sequence length, and prior type. HMM (raw) uses the time-domain pipeline without a shared deep front-end. VAE models report best-of-three prior NMI and matching macro accuracy, averaged across folds. HMM (raw) has *runs*=1.

Lab	Model	Joint NMI	Joint Acc.	Within NMI	Within Acc.
Lab 2	HMM (raw)	0.179	0.665	0.004	0.500
	cGMVAE	0.223	0.648	0.549	0.855
	HMMGMVAE	0.039	0.543	0.586	0.865
	cHMM-GMVAE	0.570	0.869	0.596	0.864
Lab 3	HMM (raw)	0.237	0.731	0.029	0.565
	cGMVAE	0.415	0.796	0.516	0.844
	HMMGMVAE	0.586	0.850	0.637	0.897
	cHMM-GMVAE	0.705	0.923	0.738	0.939
Lab 5	HMM (raw)	0.240	0.724	0.001	0.498
	cGMVAE	0.280	0.715	0.476	0.853
	HMMGMVAE	0.483	0.821	0.584	0.882
	cHMM-GMVAE	0.556	0.877	0.567	0.885

Within-lab training exceeds joint training at every site. On held-out Bern mice, cGMVAE prior NMI averages 0.22 under joint training versus 0.55 within-laboratory on the same animals. The pattern is consistent with residual covariate shift when pooling heterogeneous spectral profiles across sites. Decoder-only subject embeddings absorb amplitude nuisance during reconstruction but do not fully remove equipment and scoring context.

4.3 SUBSTAGE RESOLUTION AND PHYSIOLOGY

Expert Wake, NREM, and REM are a post-hoc reference for NMI, not a claim that sleep contains only three dynamical regimes. On holdout fold 4, macro NMI peaks near $K=3-4$ and falls at higher K , while prior log-likelihood continues to rise (Fig 1A). The same trade-off appears on the full cohort (Fig 1B). Fold 4 alone is under-powered for stable substage naming at $K \geq 6$.

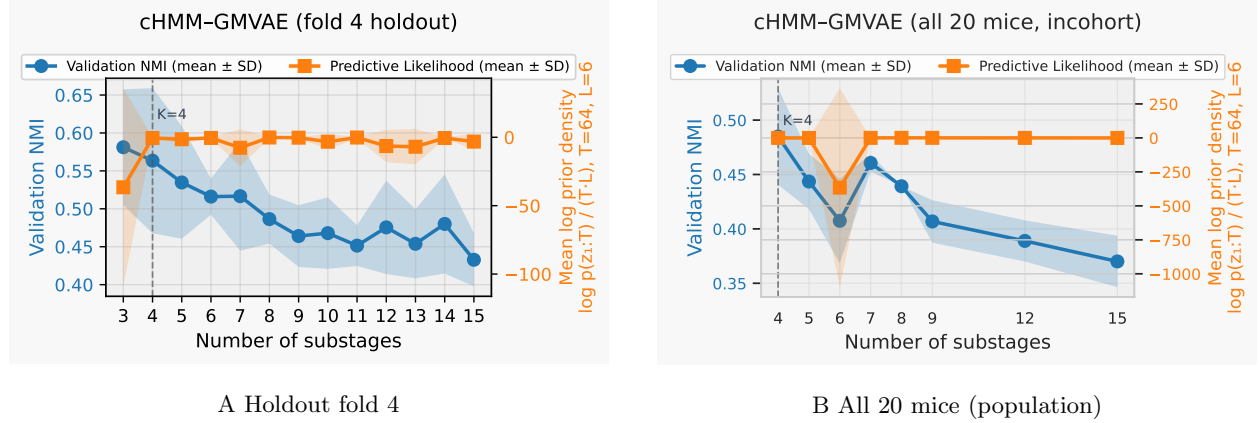


Fig 1. Substage resolution sweeps. Mean validation prior NMI (blue) and mean normalised prior log-density (orange) versus mixture count K . Shaded bands: mean $\pm$ standard deviation (SD) across seeds. A, Joint leave-mice-out holdout on fold 4 (six mice). B, Population sweep on the full cohort (same locked recipe). Macro NMI and prior density diverge once K exceeds the expert three-class grid.

We therefore interpret substage maps from a population sweep on all twenty mice under the locked recipe. Expert labels enter only for post-hoc NMI and physiology review. Fig 2 shows clearer latent separation at $K=6$ and $K=9$ than at $K=12$, with brief boundary excursions visible on the hypnogram. State distinctness in S12 Table provides a label-free backup when macro NMI falls. At population $K=6$, all six components are active with $p_{\text{perm}} = 0.02$, Cramér's $V = 0.61$, and energy distance (ED) = 2.49 despite macro NMI = 0.46. Latent substates therefore remain separated even when macro agreement falls.

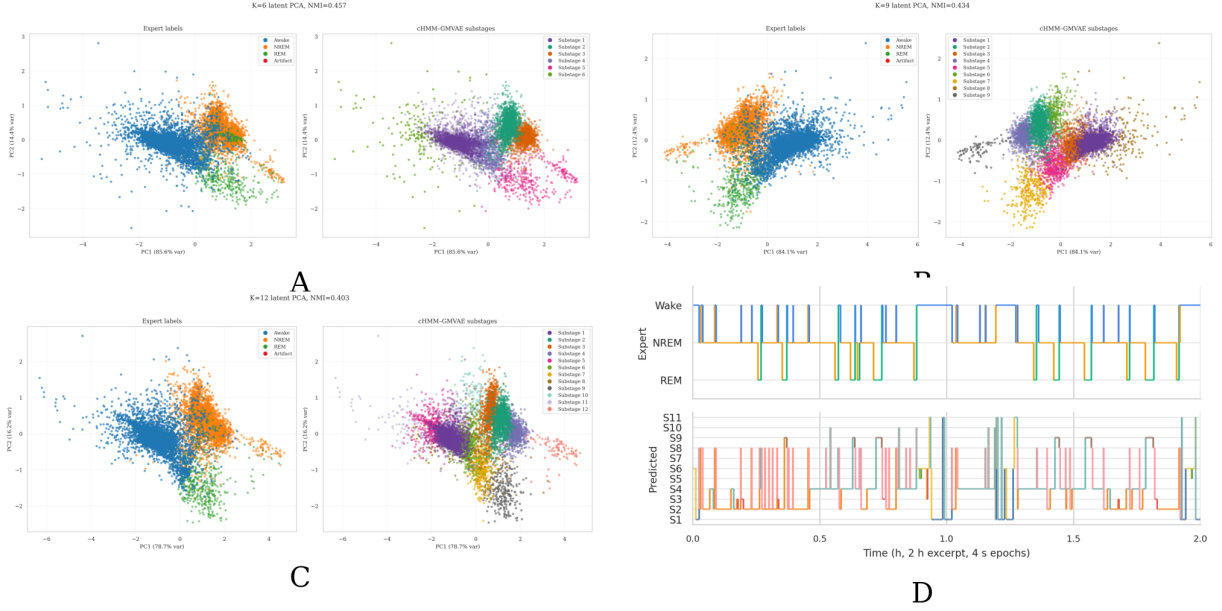


Fig 2. Latent separation and temporal structure (population sweep, best seed). A–B, principal component analysis (PCA) of encoder means: expert Wake/NREM/REM versus predicted substates at $K=6$ and $K=9$. C, Finer latent substructure at $K=12$. D, Example 30 min hypnogram at $K=12$ with macro-group background bands.

Fig 3 summarises substage physiology at $K=6$. **S2** is core quiet Wake with long bouts and about 98% expert Wake. **S5** and **S6** are dominant NREM substates with separated depth in spectra. **S3** consolidates REM at about 66% expert REM and does not split further even at $K=12$ [6,8]. **S1** and **S4** are short-bout boundary substates between Wake and NREM. Transition structure at $K=6$ is in Substage transitions at $K=6$. Together, these panels support a Wake/NREM-plus map near six substates rather than choosing K from the macro-NMI peak alone.



Fig 3. Substage physiology at $K=6$ (population sweep, best seed). Per-substage EEG/EMG power spectra, θ/δ , bout length, expert macro composition, and subject mix. Wake and NREM split into substates with distinct spectra; REM remains a single consolidated class.

4.4 LIMITATIONS

Agreement is quantified by NMI against expert macro labels, which inherit cross-laboratory disagreement [1]. REM recall on fold 4 remains near 3% despite temporal context. Cross-laboratory holdout is harder than

within-laboratory training on the same mice (Table 2). The holdout substage sweep uses fold 4 only. Population $K=6$ panels train on all twenty mice and support physiology, not leave-mice-out generalisation claims. Reported NMI uses best-of-three seeds per fold, which is optimistic relative to seed means. The method is not yet accurate enough to replace supervised staging, especially for REM.

5 CONCLUSION

Cross-laboratory mouse sleep can be staged without labels or per-mouse calibration, but only coarsely today. Temporal mixture priors improve Wake/NREM alignment on held-out mice relative to a static mixture prior, with mean prior NMI 0.575 versus 0.282 (+0.29 over folds). On the full cohort, finer structure appears at $K=6$, with six active substates separating Wake and NREM components. State distinctness remains high at $K=6$ even as macro NMI falls, which supports reading low NMI as label mismatch rather than collapsed mixtures. The approach can support comparison across acquisition sites before expert labels are harmonised. It does not yet replace supervised scoring.

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REFERENCES

1. Rose L, Zahid AN, García Ciudad J, Egebjerg C, Piilgaard L, Sørensen FL, et al. Lack of Consensus for Manual Mouse Sleep Scoring Limits Implementation of Automatic Deep Learning Models. *bioRxiv*. 2026. Available from: <https://www.biorxiv.org/content/10.64898/2026.03.27.714381v1>. doi:10.64898/2026.03.27.714381.
2. Miladinović S, Muheim C, Bauer S, et al. SPINDLE: End-to-end learning from EEG/EMG to extrapolate animal sleep scoring across experimental settings, labs and species. *PLOS Computational Biology*. 2019;15(4):e1006968. doi:10.1371/journal.pcbi.1006968.
3. Wang J, Li X, Zhang W, Chen H, Liu Y, et al. REST: A deep learning tool for automated mouse sleep stage classification. *Sleep Advances*. 2026;7(2):zpag044. doi:10.1093/sleepadvances/zpag044.
4. Barger Z, Frye CG, Liu D, Dan Y, Bouchard KE. Robust, automated sleep scoring by a compact neural network with distributional shift correction. *PLOS ONE*. 2019;14(12):e0224642. doi:10.1371/journal.pone.0224642.
5. Yaghoubi F, Sunderam S. SegWay: A novel sleep stage scoring system for mice and rats. *MethodsX*. 2016;3:369-75. doi:10.1016/j.mex.2016.03.009.
6. Katsageorgiou VM, Sona D, Zanotto M, Lassi G, Garcia-Garcia C, Tucci V, et al. A novel unsupervised analysis of electrophysiological signals reveals new sleep substages in mice. *PLOS Biology*. 2018;16(5):e2003663. doi:10.1371/journal.pbio.2003663.
7. Stevner ABA, Vidaurre D, Cabral J, Rapuano K, Nielsen SFV, Tagliazucchi E, et al. Discovery of key whole-brain transitions and dynamics during human wakefulness and non-REM sleep. *Nature Communications*. 2019;10(1):1035. Available from: <https://www.nature.com/articles/s41467-019-08934-3>. doi:10.1038/s41467-019-08934-3.
8. Elmgreen OR, Bigom AH. Characterizing sleep patterns by use of dynamic unsupervised machine learning approaches. MSc thesis, Technical University of Denmark. 2026. Population cGMVAE NMI $\approx$ 0.70; expert-feature HMM $\approx$ 0.51 (single-lab LOSO); time-domain HMM $\approx$ 0.01.
9. Rose L, Zahid AN, Ciudad JG, et al. Mouse sleep staging validation dataset (MSSV). OpenNeuro. 2025. Dataset ds006366; cite full author list from dataset paper.

10. Sunagawa G, Seo M, Sato C, et al. Fasting promotes sleep through orexin neurons. *Genes to Cells*. 2013;18(11):1107-17. FASTER staging tool lineage.
11. Yamada RG, Matsuzawa K, Ode KL, Ueda HR. An automated sleep staging tool based on simple statistical features of mice electroencephalography (EEG) and electromyography (EMG) data. *European Journal of Neuroscience*. 2024;60(7):5467-86. doi:10.1111/ejn.16465.
12. Cusinato R, Gross CT, Vyazovskiy VV, et al. Workflow for unsupervised classification of sleep states (WUCSS) in mice. *Journal of Neuroscience Methods*. 2024. doi:10.1016/j.jneumeth.2024.110156.
13. García Ciudad J, Mørup M, Kornum BR, Zahid AN. Evaluating the influence of temporal context on automatic mouse sleep staging through the application of human models. In: 2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC); 2024. Available from: <https://arxiv.org/abs/2406.16911>. arXiv:2406.16911. doi:10.1109/EMBC53108.2024.10783224.
14. Dilokthanakul N, Mediano PAM, Garnelo M, et al. Deep unsupervised clustering with Gaussian mixture variational autoencoders. In: arXiv:1611.02648; 2016. .
15. Sohn K, Lee H, Yan X. Learning structured output representation using deep conditional generative models. In: *Advances in Neural Information Processing Systems*. vol. 28; 2015. .
16. Johnson MJ, Duvenaud DK, Wiltchko A, Datta SR, Adams RP. Composing graphical models with neural networks for structured representations and fast inference. In: *Advances in Neural Information Processing Systems*. vol. 29; 2016. .
17. Ebbers J, Heymann J, Drude L, et al. Hidden markov model variational autoencoder for acoustic unit discovery. In: *Interspeech*; 2017. p. 488-92.

A DATA AND COHORT

A.1 MSSV COHORT AND QUALITY SCREENING

MSSV (OpenNeuro ds006366) [9] comprises invasive EEG and neck EMG from 92 mice across five European laboratories, with expert Wake, NREM, REM, and Artifact labels on a 4s grid for evaluation only. This report analyses laboratories 2 (Bern), 3 (Copenhagen), and 5 (Lyon): healthy wild-type sites with different electrode layouts, excluding laboratories 1 and 4 (pathology-specific or single-electrode cohorts).

Quality screening retains mice with low EEG drift and EMG that separates Wake from NREM. Drift is quantified as relative 0.5–2 Hz power ≤ 0.2 of 0.5–30 Hz on each retained electrode. Wake/NREM separability uses a root-mean-square ratio ≥ 2 on neck EMG. The screened cohort comprises 20 mice (6 Bern, 10 Copenhagen, 4 Lyon) with 42 recording runs and ~ 453 h of EEG/EMG ($\sim 4.1 \times 10^5$ artefact-free 4s epochs after screening). Training mice never appear in the validation fold for a given split (). No new animal procedures were performed.

S8 Table.

Quality-screened cohort inventory. Per-mouse recording runs, hours, and 4s epoch counts (MSSV annotation grid).

Mouse	Site	Runs	Hours	Epochs (4s)
sub-071	Bern (2)	2	12	10,800
sub-072	Bern (2)	2	12	10,800
sub-076	Bern (2)	2	12	10,800
sub-077	Bern (2)	2	13	11,700
sub-080	Bern (2)	2	12	10,800
sub-081	Bern (2)	2	12	10,800
sub-038	Copenhagen (3)	2	48	43,201
sub-039	Copenhagen (3)	3	12	10,836
sub-041	Copenhagen (3)	3	12	10,803
sub-043	Copenhagen (3)	1	24	21,600
sub-048	Copenhagen (3)	3	12	10,804
sub-054	Copenhagen (3)	2	48	43,194
sub-056	Copenhagen (3)	1	23.6	21,196
sub-059	Copenhagen (3)	1	24	21,600
sub-060	Copenhagen (3)	1	8	7,212
sub-069	Copenhagen (3)	1	24	21,600
sub-087	Lyon (5)	6	72	64,788
sub-088	Lyon (5)	2	24	21,595
sub-089	Lyon (5)	2	24	21,596
sub-092	Lyon (5)	2	24	21,595
Total (20 mice)		42	452.6	407,320

A.2 PREPROCESSING AND SPECTRAL INPUTS

Montages are harmonised to a parietal-frontal EEG differential plus neck EMG (laboratory 2: EEG1 and EEG4; laboratories 3 and 5: EEG1 and EEG2). Recordings are segmented into 4s epochs at 128 Hz. Artifact epochs are removed. Each epoch is a log-power spectral map $\mathbf{x}_t \in \mathbb{R}^{C \times F}$, batched into contiguous sequences for temporal models without hand-crafted expert features.

Per-laboratory channel sets were fixed from incohort ablation winners (full sweep outcomes in Incohort performance ceiling). Spectral passbands are summarised in Spectral bands and frequency content.

A.3 SPECTRAL BANDS AND FREQUENCY CONTENT

Each 4s epoch comprises 512 samples at 128 Hz sampling rate. The CNN input is a log-power spectrum from an unwrapped real FFT (frequency resolution 0.25 Hz; Nyquist 64 Hz). Band limiting is applied in the frequency domain on the complex spectrum before log-power: a null lower cutoff retains all bins from DC upward, and an upper cutoff zeroes bins above the configured limit. The model therefore receives the full spectral shape inside each passband, not a small vector of hand-defined sleep bands.

Locked per-laboratory passbands (holdout protocol).

Site	EEG channels	EEG passband	EMG passband
Bern (lab 2)	EEG1, EEG4	0–30 Hz	3–100 Hz (≤ 64 Hz Nyquist)
Copenhagen (lab 3)	EEG1, EEG2	0–20 Hz	5–60 Hz
Lyon (lab 5)	EEG1, EEG2	0–20 Hz	5–60 Hz

Diagnostic and physiology bands. Quality screening compares relative 0.5–2 Hz EEG power to 0.5–30 Hz total (MSSV cohort and quality screening). For substage physiology panels (Figs 2–3) and summary tables, log-power spectra are aggregated into interpretable bands on the same frequency grid: delta 0.5–4 Hz, theta 4–8 Hz, and broadband EMG power summed over the retained EMG passband. Reported θ/δ ratios use these bands.

Bern uses no per-epoch post-normalisation on log-power spectra. Copenhagen and Lyon apply per-epoch z-scoring of log-power after band limiting (post-normalisation). These limits were chosen from incohort ablations (Incohort performance ceiling) and held fixed for all cross-laboratory holdout runs.

A.4 CROSS-VALIDATION PROTOCOL

Folds were designed so each site contributes holdout mice every fold while balancing validation load: fold 1 is epoch-heaviest ($\sim 32\%$ of cohort epochs in validation, driven by long Copenhagen and Lyon recordings) whereas folds 2–4 hold out 21–25% of epochs each. sub-069 is assigned to fold 2 rather than fold 4 so Copenhagen does not contribute four simultaneous holdout mice on the substage-analysis fold. Each VAE configuration trains three random seeds per fold. Reported metrics use the best prior NMI among seeds.

S9 Table. Four-fold leave-mice-out assignments. Top: held-out mice per site and fold. Bottom: validation versus training pool sizes (joint scope trains on all non-held-out mice across sites).

Fold	Site	Held-out mice	Runs	Hours	Epochs
1	Bern (2)	sub-071	2	12	10,800
	Copenhagen (3)	sub-038, sub-039	5	60	54,037
	Lyon (5)	sub-087	6	72	64,788
2	Bern (2)	sub-072, sub-076	4	24	21,600
	Copenhagen (3)	sub-041, sub-048, sub-069	7	48	43,207
	Lyon (5)	sub-088	2	24	21,595
3	Bern (2)	sub-077	2	13	11,700
	Copenhagen (3)	sub-043, sub-054	3	72	64,794
	Lyon (5)	sub-089	2	24	21,596
4	Bern (2)	sub-080, sub-081	4	24	21,600
	Copenhagen (3)	sub-056, sub-059, sub-060	3	55.6	50,008
	Lyon (5)	sub-092	2	24	21,595

Fold	Val mice	Val runs	Val hours	Val epochs	Train mice	Train epochs	Val (%)
1	4	13	144	129,625	16	277,695	31.8
2	6	13	96	86,402	14	320,918	21.2
3	4	7	109	98,090	16	309,230	24.1
4	6	9	103.5	93,203	14	314,117	22.9

B TRAINING, CHECKPOINTS, AND METRICS**B.1 LOCKED HYPERPARAMETERS**

Latent dimension $d=6$ (matched to thesis-scale mixture models), embedding dimension 8, learning rate 1.3×10^{-3} , batch size 128, 200 epochs, three random seeds per fold. Sequence length $T=64$ epochs (~ 4.3 min);

sticky self-transition weight $\kappa=0.92$. CNN channel widths [32, 64, 128, 128], kernel sizes [7, 5, 5, 3], strides 2; MLP widths [512, 256, 128] (encoder) and [128, 256, 512] (decoder). Full YAML paths: Supplementary experimental settings.

B.2 INCOHORT ABLATION RATIONALE

Per-laboratory spectral bandpass, montage, post-normalisation, and MLP width were fixed from incohort sweeps before any holdout run (Preprocessing and spectral inputs, Incohort performance ceiling). These runs pool all mice within a site and estimate a performance *ceiling*; they do not use leave-mice-out folds.

B.3 SENSITIVITY CHECKS

On joint holdout fold 4, shortening the sequence to $T=32$ without the warm sticky schedule collapsed prior NMI (best seed ~ 0.17), whereas $T=64$ with warm training reached ~ 0.65 . On laboratory 3 incohort, warm-started HMM-GMM beat a cold HMM-GMM start (best-of-three prior NMI 0.734 vs. 0.694); gains were smaller on laboratories 2 and 5, where simpler priors remained competitive. We did not grid-search every knob at holdout; locked settings prioritise reproducibility.

B.4 WARM PRIOR SCHEDULE (cHMM-GMVAE)

Epochs 1–17: standard Gaussian KL on the encoder. Epoch 18: K -means on encoder means initialises GMM emissions. Epochs 18–36: independent mixture KL. From epoch 37: sticky HMM-GMM transitions ramp in over 18 epochs [16, 17]. β is 0 for the first ten epochs, then ramps linearly from 0.01 to 1.0.

B.5 STAGING ON ENCODER MEANS

At holdout, emissions use μ_t rather than sampled z_t . The posterior mean can bias emission scores slightly relative to stochastic latents but reduces Monte Carlo variance in Viterbi decoding and improves reproducibility across seeds once the encoder is trained.

B.6 UNSUPERVISED CHECKPOINT SCORE

Expert labels are not used to select weights during training. Let $\tilde{\omega}(\varepsilon)$ denote validation log-likelihood (sequence-normalised negative ELBO) and $H_{\text{norm}}(\hat{Y})$ the entropy of the predicted state histogram normalised by $\log K$ ($0 = \text{collapse to one state}$, $1 = \text{uniform occupancy}$). Following the thesis composite criterion [8],

$$S(\varepsilon) = \beta_{\text{ckpt}} \tilde{\omega}(\varepsilon) + (1 - \beta_{\text{ckpt}}) H_{\text{norm}}(\hat{Y}), \quad \beta_{\text{ckpt}} = 0.9. \quad (8)$$

The epoch maximising $S(\varepsilon)$ after a 5% warmup is retained per seed. Prior NMI against expert macros is computed only *after* checkpoint selection for reporting.

B.7 MACRO-ALIGNED ACCURACY AND LABEL SWITCHING

Macro accuracy maps each predicted cluster to its dominant expert macro label (Hungarian alignment when $K=3$), then reports the fraction of epochs matching Wake/NREM/REM. Accuracy can exceed chance and diverge from NMI when $K > 3$ substates collapse onto the same macro label, when too few clusters are active, or when clusters are macro-pure but permuted. Mixture components are identifiable only up to permutation; NMI is invariant to permutations, but cluster indices differ across seeds unless post-hoc alignment is applied for tables and figures.

B.8 RAW TIME-DOMAIN HMM BASELINE

Classical HMMs on raw MSSV EEG/EMG voltages do not generalise across subjects in the MSc thesis (thesis Ch. 5). On the locked cross-lab protocol, joint HMM (raw) prior NMI is near chance (mean 0.054; fold 4 peak 0.21), motivating the spectral CNN front-end used throughout this report.

B.9 SEED AGGREGATION AND SUBSTAGE VALIDITY

Each fold trains three seeds; holdout prior NMI is evaluated at each seed’s $S(\varepsilon)$ checkpoint; the maximum across seeds is the headline fold score (Tables 1–2, S11 Table). For macro model comparison we report paired fold-level ΔNMI with bootstrap 95% CI ($B=10\,000$) and note that all four folds favour cHMM-GMVAE; formal p -values with $n=4$ folds are omitted as under-powered.

For substage sweeps, macro NMI alone is insufficient when K exceeds the expert grid. We therefore report per- K validity in S12 Table: active component count, minimum occupancy, mean latent ED, Fisher trace, Cramér’s V , prior NMI, and p_{perm} . The permutation shuffles predicted substage labels on a fixed latent subsample ($n \leq 6000$ epochs) and recomputes mean pairwise ED; p_{perm} is the one-sided fraction of null values $\geq$ the observed ED. This tests whether mixture assignments carve separated regions in latent space, independent of expert label agreement.

B.10 PRIOR LOG-LIKELIHOOD AT HOLDOUT

Holdout prior log-likelihood complements NMI when judging unsupervised fit (S14 Table). For **cGMVAE** ($T=1$), each 4s epoch is treated independently: the reported value is the mean $\log p(z_t)$ under the static GMM prior on encoder means (metrics.txt Likelihood). For **HMMGMVAE** and **cHMM-GMVAE** ($T=64$), windows of 64 contiguous epochs are scored with the HMM forward marginal: metrics.txt reports mean $\log p(z_{1:T})$ per sequence (**log p(z_{1:T})**). Because the sequence length differs, raw totals are *not* comparable across cGMVAE and sequence-prior models. The normalised column divides cGMVAE values by $L=6$ (per-latent-dimension mean log-density at $T=1$) and sequence models by $T \cdot L$ (matching the K-sweep axis in Fig 1). Higher normalised values indicate better prior density on held-out latents; they need not track NMI when expert macros misalign with mixture components.

C PRIOR MODELS AND PROTOCOL ABLATIONS

C.1 HMM-GMM PRIOR (EQUATIONS)

Each discrete latent state $s_t \in \{1, \dots, K\}$ emits a diagonal Gaussian in $\mathbb{R}^d$. The per-step emission log-density is

$$\log b_k(\mathbf{z}_t) = \log \mathcal{N}(\mathbf{z}_t \mid \mathbf{m}_k, \text{diag}(\mathbf{v}_k)). \quad (9)$$

Sticky transitions use self-transition weight $\kappa=0.92$. The marginal log-likelihood of a latent window is

$$\log p(\mathbf{z}_{1:T}) = \log \sum_{s_{1:T}} \pi_{s_1} \prod_{t=2}^T A_{s_{t-1}, s_t} \prod_{t=1}^T b_{s_t}(\mathbf{z}_t), \quad (10)$$

which enters the training KL term $\mathcal{L}_{\text{KL}}$ in Eq. (5). Warm-start phases and CNN widths are in Appendix B.4 and Appendix B.1. Controlled synthetic datasets in the MSc thesis validated MAR-HMM optimisation before MSSV experiments (thesis Ch. 3); that validation is not repeated here.

C.2 DECODER-ONLY VS ENCODER+DECODER CONDITIONING

The MSc thesis default fed subject embeddings to *both* encoder and decoder on laboratory 3 only [8]. Before holdout, we compared decoder-only and encoder+decoder paths on locked incohort recipes at all three sites ($K=3$, three seeds):

Site	Decoder-only	Encoder+decoder	Best-of-three prior NMI
Bern (lab 2)	0.593	0.578	Decoder-only
Copenhagen (lab 3)	0.737	0.621	Decoder-only
Lyon (lab 5)	0.534	0.430	Decoder-only

All three sites favoured decoder-only conditioning on incohort prior NMI. The thesis headline (~ 0.70 population NMI) used encoder+decoder on laboratory 3 with expert features and leave-one-subject-out evaluation [8], which is not directly comparable to these three-lab spectral incohort runs. Holdout fold-level encoder+decoder comparisons were not run under the locked paper protocol; the main line uses decoder-only throughout.

C.3 INCOHORT PERFORMANCE CEILING

Before cross-laboratory holdout, we ran extensive within-laboratory sweeps over preprocessing, CNN-VAE architecture, prior tier (static GMM vs. HMM-GMM vs. warm sticky HMM-GMM), and sequence length T . All mice in a laboratory were pooled for training and validation tuning (no leave-mice-out). Reported values are best-of-three prior NMI at $K=3$ with decoder-only conditioning. These runs fixed the per-site recipes in Preprocessing and spectral inputs and estimate an upper performance bound when site-specific tuning is allowed. They are *not* the locked holdout protocol in Table 2. Macro accuracy was not logged systematically in the incohort sweeps.

Bern (laboratory 2): cGMVAE reached 0.593 with EEG1+EEG4, no post-normalisation, and EEG 0–30 Hz. Temporal priors did not beat the static mixture at $T=1$ (best cHMM-GMVAE 0.571 with warm prior and 50 Hz notch). Copenhagen (laboratory 3): cGMVAE peaked at 0.737; warm cHMM-GMVAE at $T=64$ matched within 0.003 (0.734). Lyon (laboratory 5): cHMM-GMVAE with EMG-wide notch preprocessing at $T=32$ reached 0.640 versus 0.534 for cGMVAE, though only one of three seeds was healthy at that peak.

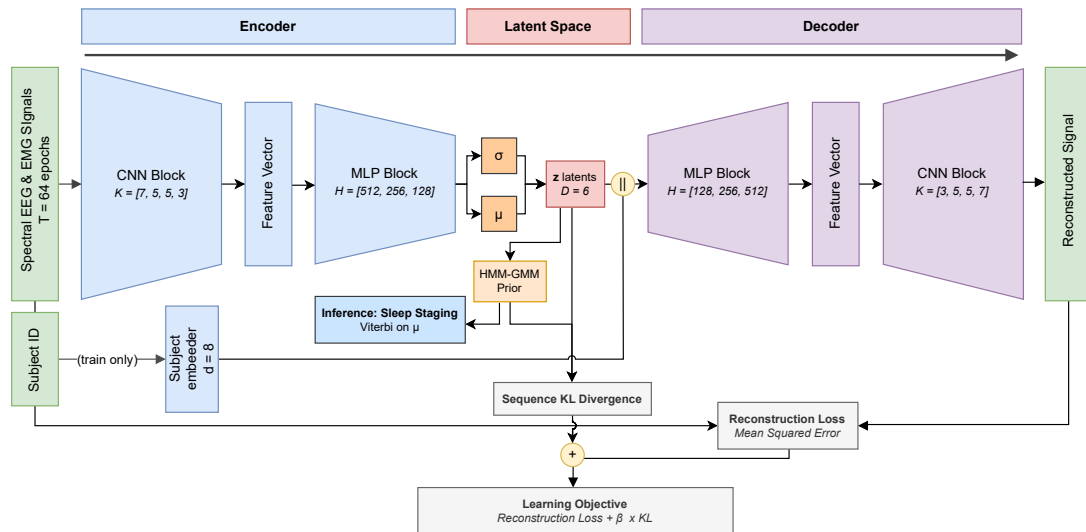
Incohort ceiling prior NMI. Best-of-three prior NMI and winning configuration per laboratory and model.

Lab	Model	Best NMI	Winning lock (summary)
2	cGMVAE	0.593	EEG1+EEG4, no post-normalisation, EEG 0–30 Hz, EMG 3–100 Hz
2	cHMM-GMVAE	0.571	warm prior + 50 Hz notch, below cGMVAE at $T=1$
3	cGMVAE	0.737	post-normalisation, EEG 0–20 Hz, MLP widths 512–256–128
3	cHMM-GMVAE	0.734	warm HMM-GMM, $T=64$, same preprocessing and architecture locks
5	cGMVAE	0.534	post-normalisation, EEG 0–20 Hz, MLP widths 512–256–128
5	cHMM-GMVAE	0.640	EMG wide + notch, $T=32$ (1/3 seeds healthy)

Ceiling reference only. Holdout within-lab NMI and accuracy under the same locked recipes appear in Table 2. Joint holdout remains below these ceilings because leave-mice-out validation is strictly harder than pooling all laboratory mice during tuning.

D SUPPLEMENTARY FIGURES

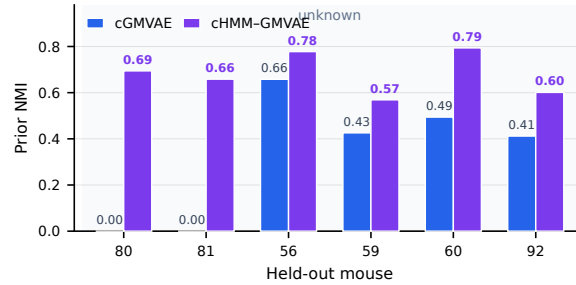
D.1 MODEL ARCHITECTURE



Conditional Gaussian mixture variational autoencoder (GMVAE) architecture. Encoder: spectral EEG/EMG maps pass through CNN ($K=[7, 5, 5, 3]$) and MLP blocks. Subject ID feeds a subject embedder used only

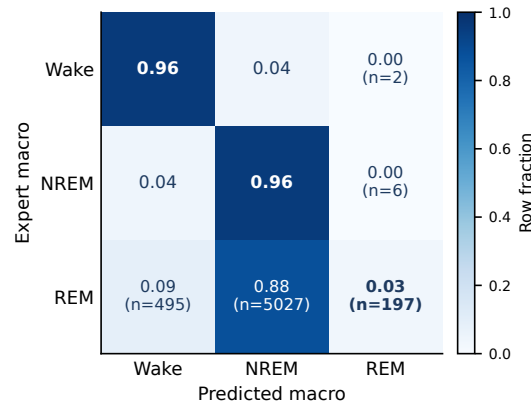
in the decoder. Latent: diagonal Gaussian posterior with dimension $d=6$ feeds a GMM (cGMVAE) or HMM-GMM (cHMM-GMVAE) prior. Decoder: latent vector concatenated with subject embedding, then CNN reconstruction. Training minimises mean squared error (MSE) reconstruction plus $\beta \times$ prior KL. At holdout, only the encoder and prior are used for staging.

D.2 S2 FIG.

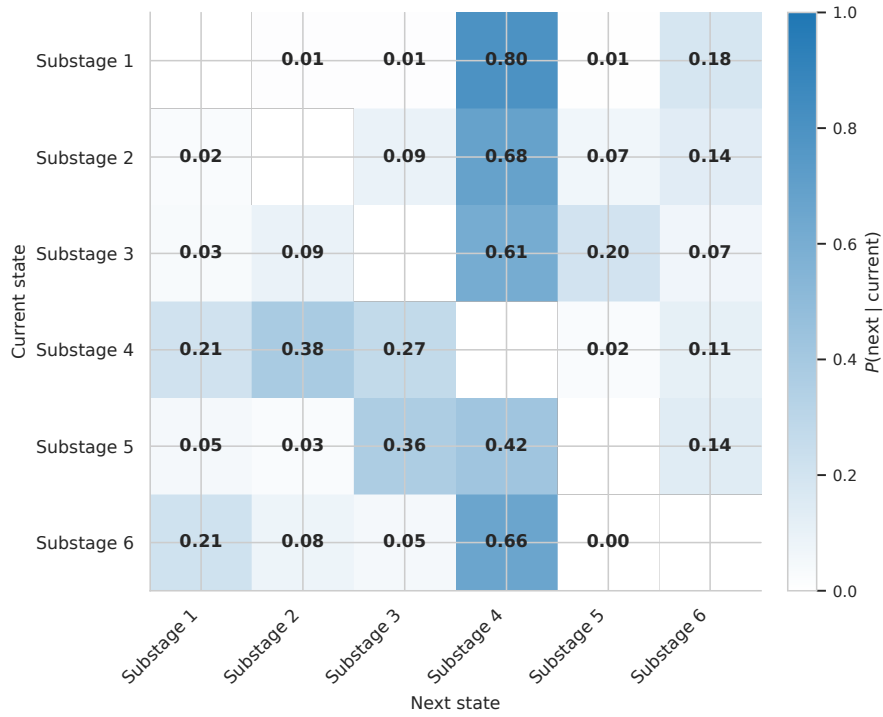


Per-mouse prior NMI on fold 4. Paired bars for cGMVAE and cHMM-GMVAE on each held-out mouse.

D.3 S3 FIG.



Macro confusion (fold 4, $K=3$, cHMM-GMVAE, best seed). Row-normalised 3×3 matrix: expert Wake/NREM/REM (rows) vs. predicted macro after Hungarian alignment (columns). REM row shows low recall ($\sim 3\%$) on this holdout fold.

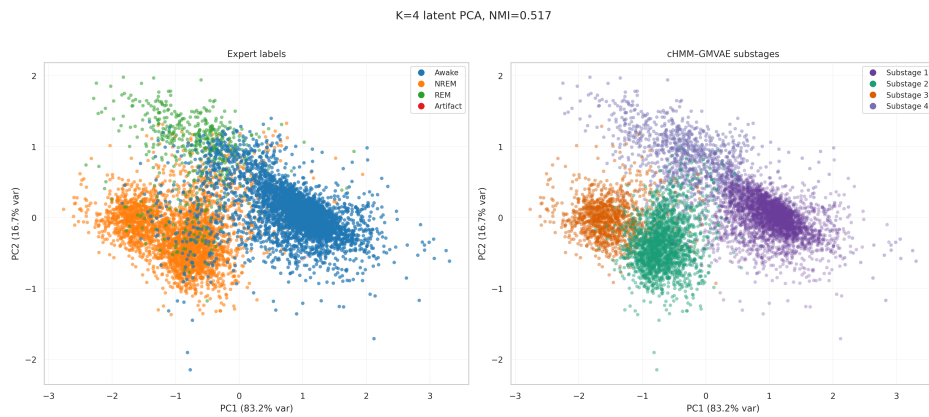
D.4 SUBSTAGE TRANSITIONS AT $K=6$ 

Substage dynamics at $K=6$ (population sweep, best seed). Change-only transition matrix $P(\text{next} | \text{current})$ between mixture components. Stable within-macro micro-cycles in Wake S2 and NREM S5/S6; short-lived boundary substates S1 and S4 mediate Wake–NREM switches.

D.5 POPULATION SUBSTAGE PCA ($K=4, 7, 8$)

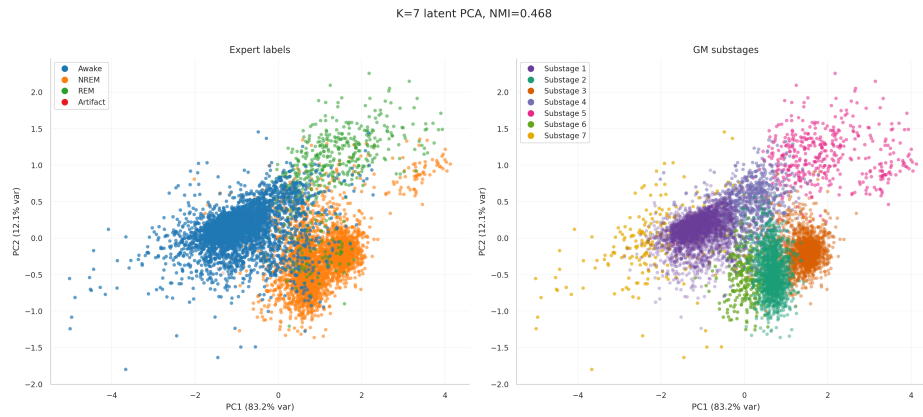
Locked cHMM–GMVAE, all 20 mice (incohort), best seed per K . Each panel shows PCA of encoder means with expert Wake/NREM/REM versus predicted substates.

D.6 S17 FIG.



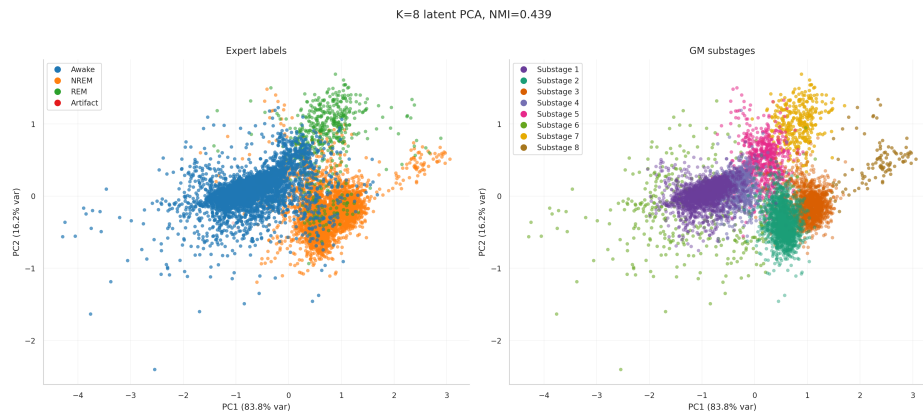
Population $K=4$ (best seed, prior NMI 0.517). Latent PCA: expert Wake/NREM/REM versus predicted substates.

D.7 S18 FIG.



Population $K=7$ (best seed, prior NMI 0.468). Latent PCA: expert macros versus predicted substates.

D.8 S19 FIG.



Population $K=8$ (best seed, prior NMI 0.439). Latent PCA: expert macros versus predicted substates.

E SUPPLEMENTARY TABLES

E.1 S10 TABLE.

Per-class classification metrics (fold 4, $K=3$, cHMM-GMVAE, best seed). Cohen's κ , macro F1, and per-class recall after Hungarian alignment.

Run	κ	Macro F1	Wake rec.	NREM rec.	REM rec.
fold4 cHMM $K=3$	0.811	0.638	0.959	0.955	0.034

E.2 S11 TABLE.

Seed stability (joint holdout). Prior NMI mean, standard deviation, and best-of-three per fold and model.

Fold	Model	Seeds	NMI mean	NMI SD	NMI best
1	cGMVAE	2	0.219	0.048	0.267
1	cHMMGMVAE	3	0.521	0.041	0.579
2	cGMVAE	2	0.168	0.089	0.256
2	cHMMGMVAE	3	0.375	0.014	0.396
3	cGMVAE	2	0.297	0.009	0.307
3	cHMMGMVAE	3	0.450	0.315	0.675
4	cGMVAE	2	0.214	0.085	0.299
4	cHMMGMVAE	3	0.581	0.056	0.649

E.3 S12 TABLE.

Substage validity across mixture resolution (population sweep, best seed per K). Active: mixture components with ≥ 1 assigned epoch. Min occ.: occupancy of the rarest active substage. Latent ED: mean pairwise energy distance between substates in encoder space. Fisher: Fisher trace of latent between-state scatter. Cramér V : association between predicted substages and expert macros (post-hoc). p_{perm} : permutation test for latent separation (shuffle substage labels; H_0 : observed ED is label-random). Low p_{perm} supports distinct substates even when macro NMI falls at high K .

K	Active	Min occ. (%)	Latent ED	Fisher	Cramér V	NMI	p_{perm}
4	4/4	0.37	1.91	2.14	0.503	0.542	0.010
5	5/5	6.19	2.61	5.98	0.591	0.484	0.010
6	6/6	3.13	2.46	7.19	0.609	0.457	0.010
7	7/7	3.19	2.50	9.10	0.631	0.468	0.010
8	8/8	1.38	2.20	9.62	0.659	0.439	0.010
9	9/9	1.35	2.68	11.36	0.675	0.442	0.010
12	12/12	1.38	3.03	11.68	0.662	0.403	0.010
15	10/15	0.01	3.48	5.16	0.496	0.409	0.010

E.4 S14 TABLE.

Holdout prior log-likelihood (joint scope, $K=6$, best-of-three seed). cGMVAE uses sequence length $T=1$: each 4s epoch is scored independently with the static GMM prior (mean $\log p(z_t)$ per epoch; **Likelihood** in run metrics). HMMGMVAE and cHMM-GMVAE use $T=64$ contiguous epochs per window; the reported value is the HMM forward marginal mean $\log p(z_{1:T})$ per sequence (**log p(z_1:T)**). Raw totals are *not* comparable across $T=1$ and $T=64$ models because sequence models accumulate density over 64 steps. The normalised column divides cGMVAE by $L=6$ and sequence models by $T \cdot L=384$, matching the orange axis in Fig 1 for $T=64$ runs. Higher values indicate better prior density on held-out encoder means; they need not track NMI when mixture structure misaligns with expert macros. Fold 4 cGMVAE (seed 2) shows an extreme negative raw value despite moderate NMI—a known case where prior density and clustering quality diverge.

Fold	Model	T	Holdout NMI	Prior LL (raw)	Norm.
1	cGMVAE	1	0.267	-17.59	-2.931
1	HMMGMVAE	64	0.409	-3755.5	-9.780
1	cHMM-GMVAE	64	0.579	-26.8	-0.070
2	cGMVAE	1	0.256	-11.86	-1.976
2	HMMGMVAE	64	0.236	-1200.4	-3.126
2	cHMM-GMVAE	64	0.396	-307.3	-0.800
3	cGMVAE	1	0.307	-12.51	-2.085
3	HMMGMVAE	64	0.497	-166.0	-0.432
3	cHMM-GMVAE	64	0.675	-84.5	-0.220
4	cGMVAE	1	0.299	-68904.82	-11484.137
4	HMMGMVAE	64	0.404	-264.6	-0.689
4	cHMM-GMVAE	64	0.649	281.1	0.732

E.5 S15 TABLE.

Exploratory substage map ($K=12$, population sweep, seed 2). Mixture index S1–S12 with occupancy, dominant expert macro, biological reading, and proposed merges after sleep-biology review (A. Wollesen Jürgensen-Breum, B. Kornum). Dominant macro is post-hoc only; training is fully unsupervised.

State	Occ. (%)	Dominant macro	Biological reading and proposed action
S1	5.3	Mixed W/N/R	Short-bout transition bucket; merge with S6 or drop
S2	3.2	REM (93%)	Consolidated REM; model does not split REM further
S3	25.7	Wake (99%)	Core quiet Wake (largest Wake cluster)
S4	9.2	Wake (99%)	Spectrally deep/slow Wake neighbour; review vs. N3-like label
S5	1.4	NREM (82%)	High EMG; 5 mice only—artefact / line noise
S6	2.3	NREM (78%)	Light NREM (N1-like); merge with S10
S7	2.0	Wake (98%)	REM→Wake transition (noisy post-REM epochs)
S8	7.6	Wake (90%)	Short-bout Wake–NREM mix; merge with neighbours
S9	4.9	Mixed W/N	Wake–NREM boundary; merge with S6/S10
S10	12.6	NREM (94%)	Core NREM (N2-like)
S11	11.5	NREM (95%)	High-variability tail of S5-like EMG noise or over-split active Wake
S12	14.3	NREM (94%)	Redundant NREM bucket; collapse into S10

E.6 SUPPLEMENTARY EXPERIMENTAL SETTINGS

Full hyperparameters for joint and within-laboratory holdout models are provided as supplementary material.